Gao Mingyang

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Education

School of Computing and Data Science, The University of Hong Kong **(expected graduation: 2027-5)**

- BSc in Statistics; Minor in Mathematics
- CGPA: 3.71 (Dean's Honours List)

Coursework

Math: Multivariable Calculus, Linear Algebra, Stochastic Process, Nonlinear Optimization, ODE, Real and Complex Analysis
Statistics: Probability Theory, Mathematical Statistics, Statistical Inference, Machine Learning, Bayesian Learning, Time Series and Forecast, Nonparametric Statistics, Intelligent Market Models
Coding: Python, R, Stan, Julia, MATLAB. Particularly experienced in Scikit-Learn, Pytorch, Optuna, and probabilistic programming using Stan and PyMC

Research Experience

Research Intern, School of Computing and Data Science, HKU **(2026-05 – 2026-09)**
Project Topic: Efficient and approximate MCMC algorithms for Gaussian Process
Supervisor: Prof. Zhu Yichen, School of Computing and Data Science, HKU

- Investigated the convergence property of an approximate Gibbs sampler for the posterior inference of a high dimensional Gaussian Process, where a conjugate gradient linear solver is used in each sampling iteration, providing theoretical guarantee for a scalable, high performance sampling scheme.
- Conducted numerical experiments using Python and Julia to empirically demonstrate the convergence of the approximate chain with the exact chain and to solve for an empirical contraction factor.

Research Assistant, School of Computing and Data Science, HKU **(2025-06 – 2026-59)**
Project Topic: Automatic Optimization of Explainable Machine-Learning Models for Predicting Insurance Recovery
Supervisor: Prof. Tim Boonen, School of Computing and Data Science, HKU

- Built a complete machine learning pipeline using disability insurance dataset from SoA; deployed sophisticated ML models such as XGBoost, Neural Network and ensemble models, significantly improved prediction performance.
- Employed SMOGN, a modern resampling method to deal with class imbalance in regression; Recoded the Python package using Julia to enable efficient computation.
- Utilized Optuna, a state-of-the-art optimization package, to conduct auto-hyperparameter search and multi-objective optimization, achieved superior results than traditional grid search and outperformed similar studies.
- Deployed SHAP to explain variable influence and importance, enabling the explanation of black-box models.

Team Lead, 30-Day Hospital Readmission Prediction, Kaggle Competition **(2025-09 – 2025-12)**

- Led a team of 4 to develop a complete pipeline from feature engineering to model optimization to predict patient readmission using multimodal medical data; implemented effective feature enhancement and hyperparameter selection that produced a top-performing model.
- Designed a CNN model to extract image features and tuned a TF-IDF tokenizer and BERT to extract text data from hospital notes; built an ensemble model using multimodal data consisting of text, image and patient logs, significantly improving ROC-AUC performance and was highly competitive among participating teams.

Publications

Predicting Recovery in Group Long-Term Disability Insurance with Optimized Explainable Machine Learning, 2026, *https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7352605*

Teaching

Undergraduate Teaching Assistant, Mathematical Statistics, STAT/SDST 2602 **(2026 Fall)**

- Responsible for tutorial demonstration, assignment grading, peer advising and test invigilation.

Extracurricular

School Peer Advisor, School of Computing and Data Science, HKU **(2025-09 – now)**

- Provided personal consultations to advisees, from academic topics like course and major selection, to daily affairs such as adapting to life at Hong Kong.

Experiential Learning Ambassador, Faculty of Science, HKU **(2024-09 - now)**

- Co-hosted conversion events for incoming students, presented to 100+ audiences about general learning and

Statistics study at HKU.

Language
English: TOEFL: 118/120

Awards
HKU Worldwide Exchange Scholarship **(2026 spring)**
Summer Research Fellowship, Faculty of Science **(2025 summer)**
Mitacs Globalink Research Internship **(Awarded, 2026 spring)**
Dean’s Honour List **(2025 summer)**
HKU China Vision Winter Internship **(2024 winter)**